Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3311-WE01

Title: (BEng/MEng)

Geotechnics 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Q1

A 2-storey building is projected on a site that has the ground profile indicated in Figure Q1a, with the water table at the ground surface. The geotechnical engineers of the project have concerns about the mechanical properties of the 10m thick layer of saturated normally consolidated clay. To understand better its compressibility, they decide to take some undisturbed clay samples to the soil mechanics lab, all taken from point A which is at a depth of 2.5m from the ground surface (see point A in Figure Q1a). The bulk unit weight of the saturated normally consolidated clay is 19.5 kN/m^3.

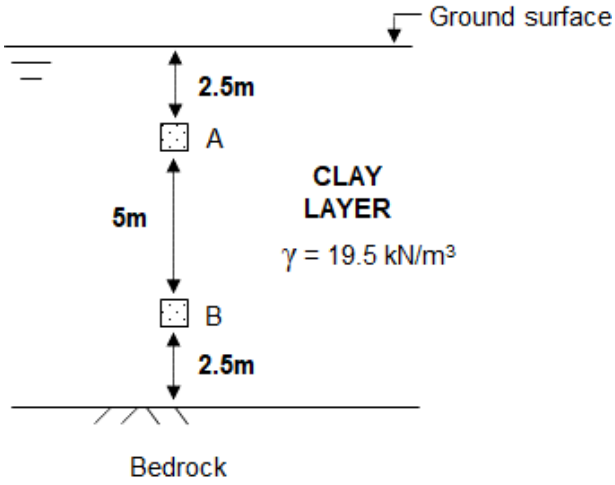


Figure Q1a (Not to scale)

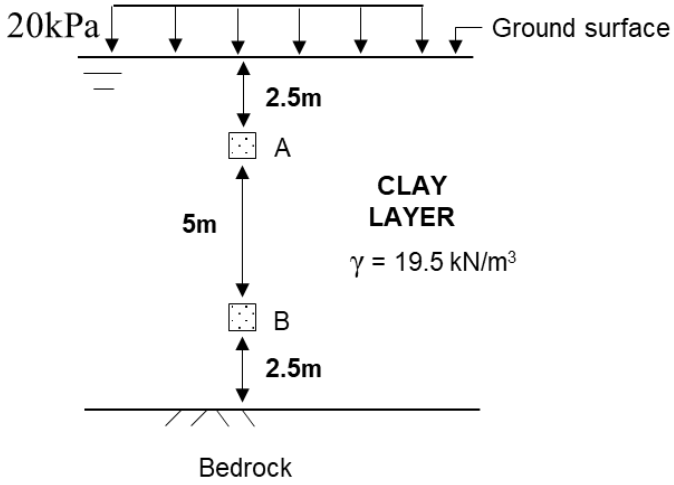


Figure Q1b (Not to scale)

(a) A small mass of 270.2g of the soil sampled is used to determine its gravimetric water content. If after oven drying, the dry mass is 210.5g, calculate the in-situ gravimetric water content at A. **[15%]**

(b) Assuming that the specific gravity of the grains is 2.67, calculate the in-situ void ratio at A. **[15%]**

(c) An undisturbed cylindrical specimen from the remaining soil sampled is tested under oedometric conditions. Estimate the value of the void ratio in-situ using information of its compression index (C_c = 0.45) and knowing that the value of the intercept of the one-dimensional (1D) normal compression line (at a value of vertical effective stress of 1 kPa) is 1.375. **[20%]**

(d) Under the assumptions that the soil at A and at B (see Figure Q1a) has been only subjected to 1D straining, calculate the in-situ void ratio at B. **[20%]**

(e) If the friction angle of the normally consolidated clay is 13º, calculate the values of the horizontal effective stress at A and at B. **[20%]**

(f) To explore the feasibility of a very wide foundation to support the 20kPa of uniform pressure caused by the weight of the 2-storey building (see Figure Q1b), estimate the total settlement. Justify whether this type of foundation is a sensible option for this project. **[10%]**

continued

Q2

A retaining gravity wall (with a unit weight of 25kN/m^3) is to be constructed to provide stability in a vertical cut of a 5m excavation, as indicated in Figure Q2. The ground excavated is a silty clay with a bulk unit weight of 20 kN/m^3. A drained system is design so that any horizontal water pressures acting on the wall can be disregarded.

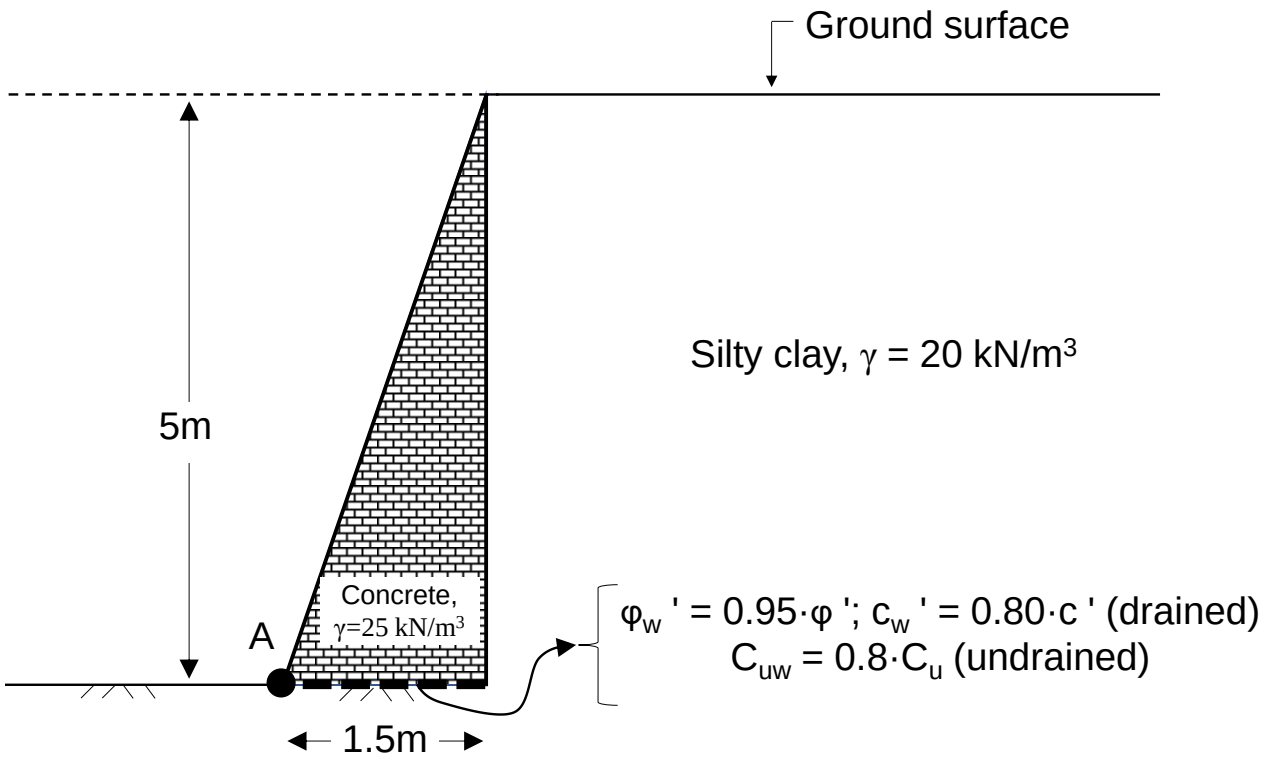


Figure Q2 (Not to scale)

To study the safety of this geotechnical structure, two hypotheses are investigated: (i) failure under drained conditions at the long term and considering a dry soil condition (with c' = 10 kPa and φ' = 30º) and (ii) failure under undrained conditions at the short term and considering the soil saturated (C_u = 20 kPa). Both hypotheses assume Rankine's approach.

(a) Sketch the active pressures diagram on the wall in each of these hypotheses, indicating the relevant values of active pressures and positions in the diagram. **[30%]**

(b) Calculate the total horizontal thrust and its position for both failure cases considered. Calculate also the weight force of retaining structure and indicate its position. **[20%]**

(c) Calculate the conventional factor of safety (i.e., with no consideration of partial factors) for overturning about A and for sliding. Consider that friction between the ground and the concrete is only relevant at the base (not behind the wall) and it can be approximated by the Mohr-Coulomb failure criterion with $c_w = 0.8 \cdot c'$ and $\varphi_w' = 0.95 \cdot \varphi'$ (for the drained case) and by the Tresca failure criterion with $C_{uw} = 0.8 \cdot C_u$ (for the undrained case). **[20%]**

(d) Study the influence of c' and C_u in the factor of safety considering c' ± 10kPa and C_u ± 10 kPa. **[30%]**

continued

Q3

The design of a square footing on a 10m thick layer of clay with the water table on the ground surface is being investigated (see Figure Q3). The lab report of the mechanical properties of this saturated clay informs that its friction angle is 25°, its cohesion is 30 kPa, its undrained shear strength is 70 kPa and its saturated unit weight of 20kN/m^3. The footing is meant to support a telecommunications tower which transmits a load of V = 10,000kN.

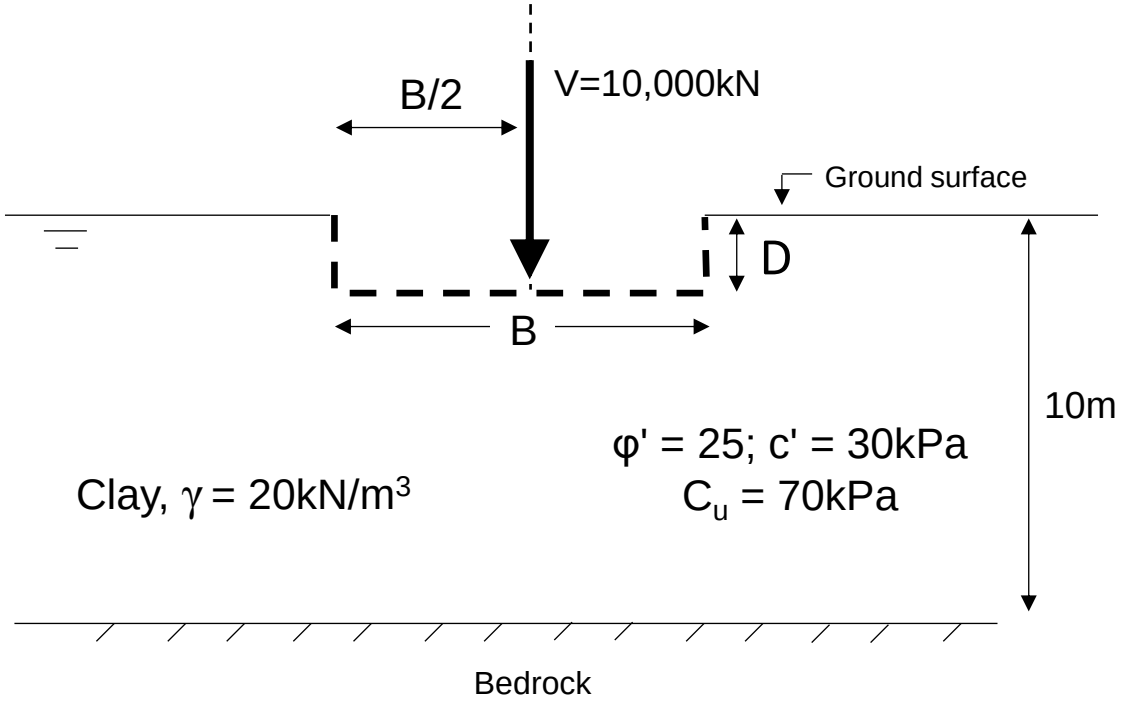


Figure Q3

A conventional design analysis (i.e., with no partial factors) is proposed to determine the dimensions of the footing for the short-term (undrained) behaviour. Consider that the required factor of safety is at least 3 and ignore any depth factor.

(a) Calculate the plan dimensions of the footing for D = 0m. **[20%]**

(b) Based on the above considerations, would you suggest carrying out also the analysis for the long-term (drained) behaviour? Justify your answer. **[10%]**

(c) To analyse the influence of the depth D at which the footing will rest, calculate the plan dimensions of the footing for D = 1m and D = 2m in the short-term. Consider that the bearing capacity can be estimated as $q_f = 5.14 \cdot C_u \cdot s_c + p_{ov}$ where p_{ov} is the overburden pressure. **[20%]**

(d) How does D influence the results? Justify your answer. **[10%]**

(e) If the vertical load V is now inclined 5° with respect to the vertical direction (see dashed line in Figure Q3), calculate the value of B for the short-term conditions and D = 0, 1 and 2m (assume that all inclination load factors are one). **[20%]**

continued

(f) Under the scenario of an inclined load mentioned in (e), should the geotechnical design include another safety check? Justify your answer. **[20%]**

Phase relationships

$e=\frac{V_v}{V_s}$	$S_r=\frac{V_w}{V_v}$	$v=1+e$	$n=\frac{e}{1+e}$	$w=\frac{M_w}{M_s}$
$\rho_s=G_s\rho_w$	$wG_s=S_re$	$\rho=\frac{M_s+M_w}{V_s+V_v}$	$\rho=\frac{(G_s+S_re)\rho_w}{1+e}$	$\gamma=\rho g$
$\gamma=\frac{(G_s+S_re)\gamma_w}{1+e}$	$\gamma'=\gamma_{sat}-\gamma_w$	$\rho_d=\frac{G_s\rho_w}{1+e}$	$\rho_d=\frac{\rho}{1+w}$	$\gamma_w\approx 10\ \text{kN/m}^3$

Stresses and pore pressures in soils:

$\sigma_v=\rho gz=\gamma z$	$u_w=\rho_w gz_w=\gamma_w z_w$	$\sigma_v'=\sigma_v-u$	$u_a-u_w=\frac{2T}{R}$	$h_c=\frac{2T}{\gamma_w R}$

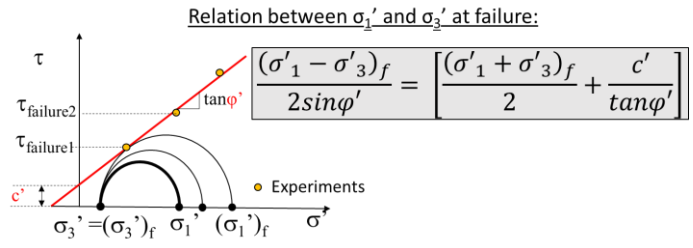
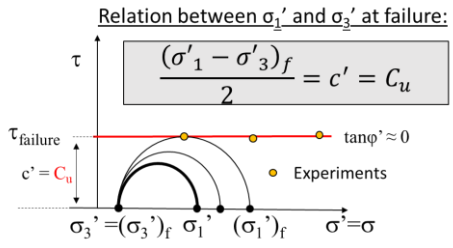
Elastic stress-strain relations:

The volumetric strain is: $\Delta\varepsilon_v=\Delta\varepsilon_1+\Delta\varepsilon_2+\Delta\varepsilon_3=\Delta\varepsilon_x+\Delta\varepsilon_y+\Delta\varepsilon_z$

$\Delta\varepsilon_x=\frac{1}{E'}\left(\Delta\sigma_x'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{xy}=\frac{\Delta\tau_{xy}}{G'}$	--
$\Delta\varepsilon_y=\frac{1}{E'}\left(\Delta\sigma_y'-\nu'\Delta\sigma_x'-\nu'\Delta\sigma_z'\right)$	$\Delta\gamma_{yz}=\frac{\Delta\tau_{yz}}{G'}$	$K'=\frac{E'}{3(1-2\nu')}$
$\Delta\varepsilon_z=\frac{1}{E'}\left(\Delta\sigma_z'-\nu'\Delta\sigma_y'-\nu'\Delta\sigma_x'\right)$	$\Delta\gamma_{zx}=\frac{\Delta\tau_{zx}}{G'}$	$G'=\frac{E'}{2(1+\nu')}$
$\Delta\varepsilon_v=\frac{\Delta p'}{K'}$		$\Delta\varepsilon_q^e=\frac{\Delta q}{3G'}$
$p'=\frac{\sigma_1'+\sigma_2'+\sigma_3'}{3}$		$q=\left[\frac{(\sigma_1'-\sigma_2')^2+(\sigma_2'-\sigma_3')^2+(\sigma_3'-\sigma_1')^2}{2}\right]^{1/2}$

where σ_1', σ_2' and σ_3' are the principal effective stresses (i.e. $\sigma_1'\geq\sigma_2'\geq\sigma_3'$)

Failure envelopes:

$\tau=c'+\sigma'\tan\varphi'$ (Mohr-Coulomb)	$\tau=C_u$ (Tresca)
Relation between σ_1' and σ_3' at failure: 	Relation between σ_1' and σ_3' at failure: 

Ground water flow:

$h=z+\frac{u}{\rho g}=z+\frac{u}{\gamma_w}$	$q_x=Aki_x=Ak\left(-\frac{\partial h}{\partial x}\right)$	$v_x=ki_x$	$v_{true,x}=v_x\frac{1}{n}$

Constant head permeameter:	Falling head permeameter:	Pumping test:
$q=Ak\frac{h_1-h_2}{L}$	$\ln\left(\frac{h_0}{h}\right)=\frac{Ak}{aL}t$	$\left(H^2-h^2\right)=\frac{q}{\pi k}\left(\ln R_0-\ln r\right)$

Laplace's Equation (3D)	Laplace's Equation (2D, flownets)	
$\frac{\partial^2 h}{\partial x^2}+\frac{\partial^2 h}{\partial y^2}+\frac{\partial^2 h}{\partial z^2}=0$	$\delta h=\frac{\Delta h}{N_d}$	$q=k\frac{N_f}{N_d}\Delta h;\ \ u=\gamma_w\left(h-z\right)$

continued

Critical hydraulic gradient:	$i_c = \dfrac{\gamma - \gamma_w}{\gamma_w}$

1D flow through an unconfined aquifer:	1D flow through a confined aquifer:
$q_x = \dfrac{k\left(H_1^2 - H_2^2\right)}{2L}$	$q_x = Aki_x = -Ak\dfrac{\partial h}{\partial x} = -k\dfrac{H_2 - H_1}{L}D$
$h = \sqrt{\dfrac{\left(H_2^2 - H_1^2\right)}{L}x + H_1^2}$	$h = \dfrac{H_2 - H_1}{L}x + H_1$

1D straining (oedometric test conditions):

1D normal compression line:	1D swelling line:
$e = e_1 - C_c \log_{10} \sigma'_v$	$e = e_2 - C_s \log_{10} \sigma'_v$

$OCR_{1D} = \dfrac{\textit{max. past value of } \sigma'_v}{\textit{current value of } \sigma'_v} = \dfrac{\sigma'_{vc}}{\sigma'_v}$	$\sigma'_h = K_0 \sigma'_v$	$K_{0nc} \approx 1 - \sin\varphi'$	$K_0 \approx (1 - \sin\varphi')\sqrt{OCR}$

Settlement of a very wide foundation:		
$\dfrac{\Delta H}{H_0} = \dfrac{e_0 - e_1}{1 + e_0}$	$\Delta H = \dfrac{H_0}{1 + e_0}\left[C_s \log_{10}\dfrac{\sigma'_{vc}}{\sigma'_{v0}} + C_c \log_{10}\dfrac{\sigma'_{v1}}{\sigma'_{vc}}\right]$	$\Delta\varepsilon_v = m_v \Delta\sigma'_v$

1D consolidation:		
$\dfrac{\partial u_e}{\partial t} = c_v\left(\dfrac{\partial^2 u_e}{\partial z^2}\right)$	$c_v = \dfrac{k}{m_v \gamma_w}$	$T_v = \dfrac{c_v t}{d^2}$

Axisymmetric conditions (triaxial test conditions):

$\Delta\varepsilon_v = \Delta\varepsilon_a + 2\Delta\varepsilon_r$	$\Delta\varepsilon_q = \dfrac{2}{3}\left(\Delta\varepsilon_a - \Delta\varepsilon_r\right)$
$p' = \dfrac{1}{3}\left(\sigma_a' + 2\sigma_r'\right)$	$q = \sigma_a - \sigma_r$

where σ'_a and σ'_r are the principal effective axial and radial stresses respectively

Iso-normal compression line:	Iso-swelling line:
$v = N - \lambda \ln p'$	$v = v_\kappa - \kappa \ln p'$

Critical states:	
$q = Mp'$	$v = \Gamma - \lambda \ln p'$

where Γ=N-(λ-κ)ln(2);

Modified Cam Clay yield curve *f*:		$q^2 - M^2 p'(p'_0 - p') = 0$	
Plastic volumetric strains	$\Delta\varepsilon_v^p = \dfrac{(\lambda - \kappa)}{v} ln\left(\dfrac{p'_{0B}}{p'_{0A}}\right)$	Plastic deviatoric strains	$\Delta\varepsilon_q^p = \dfrac{2\eta}{M^2 - \eta^2}\Delta\varepsilon_v^p$
Stress ratio		$\eta = \dfrac{q}{p'}$	
Relation between friction angle and critical state gradient:		$M = \left.\dfrac{q}{p'}\right\|_{\textit{at critical state}} = \dfrac{6\sin\varphi'}{3 - \sin\varphi'}$	

continued

Overconsolidation ratio (OCR in terms of p')	$OCR = \frac{max.\ past\ value\ of\ p'}{current\ value\ of\ p'} = \frac{p'_0}{p'}$

Partial factors Eurocode 7 & factor of safety: (Bond & Harris, 2008; *Decoding EC7*)

$q_{Ed} \le q_{Rd}$ where, q_{Ed} is the design pressure and q_{Rd} is the design resistance	CASE	ACTIONS			Soil properties	
		Permanent		Variable	tan φ', c'	C_u
		Unfavourable	Favourable	Unfavourable		
	C	1.00	1.00	1.30	1.25	1.4

Factor of safety, FoS:	$FoS = \frac{available\ shear\ force,\ T_f}{allowable\ working\ load,\ T}$; $FoS = \frac{available\ bearing\ capcity,\ \tau_f}{allowable\ bearing\ capacity,\ \tau}$

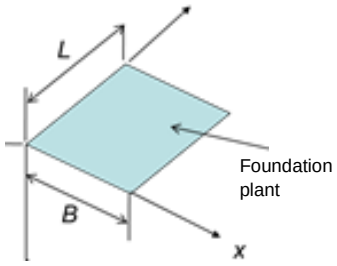
Shallow foundation settlement, δ:

Cohesionless soils:

Burland & Burbage:

$$\delta\ (mm) = f_s \cdot f_l \cdot f_t \cdot \left[\left(q'_n - \frac{2}{3}p'_0\right) \cdot B^{0.7} \cdot I_c\right]$$

where,



$f_s = \left(\frac{1.25 \cdot L/B}{\frac{L}{B} + 0.25}\right)^2$	$f_l = \frac{H}{z_l}\left(2 - \frac{H}{z_l}\right)$	$f_t = 1 + R_3 + R \cdot log\left(\frac{t}{3}\right)$

Cohesive soils: $\delta = \delta_i + \delta_c$ where,

$\delta_i\ (mm) = \mu_1 \cdot \mu_0 \cdot q_n \frac{B}{E_u}$	$\delta_c\ (mm) = \mu_g \cdot \delta_{oed}$ where, $\delta_{oed}\ (mm) = m_v \cdot \sigma'_v \cdot H$

Bearing capacity:

Brinch-Hansen (drained all soils)

$$q_f = c' \cdot N_c \cdot s_c + p'_{ov} \cdot N_q \cdot s_q + \frac{1}{2} \cdot \gamma' \cdot B \cdot N_y \cdot s_y$$

where,

$N_q = e^{\pi \tan \varphi'} \cdot \left(\tan(45 + \varphi'/2)\right)^2$	$s_q = 1 + (B'/L') \cdot \sin \varphi'$ (rectangular); $s_q = 1 + \sin \varphi'$ (circular and square)
$N_c = \left(N_q - 1\right) \cdot \cot \varphi'$	$s_c = \left(s_q \cdot N_q - 1\right)/\left(N_q - 1\right)$ (all)
$N_y = 2 \cdot \left(N_q - 1\right) \cdot \tan \varphi'$	$s_y = 1 - 0.3 \cdot (B'/L')$ (rectangular; $s_y = 0.7$ (circular and square)

depth factors are not permitted within EC7; load inclination, base and ground factors ignored.

Fine-grained soils (undrained clays; less accurate approximation if used for sands and gravels)

$q_f = 5.14 \cdot C_u \cdot s'_c + p'_{ov}$ *where,* $s'_c = 1 + 0.2 \cdot \left(\frac{B'}{L'}\right)$ *(rectangular shape; for circular and square shape B'=L'); Note: If no overburden is considered,* $q_{f(net)} = 5.14 \cdot C_u \cdot s'_c$

load inclination, base and ground factors are ignored.

Bearing capacity in piles:

$$Q_u = Q_s + Q_b - W_p$$

Cohesionless soils:	Cohesive soils:
$Q_s = q_s \cdot A_s = A_s \cdot k_s \cdot \sigma_{v0} \cdot \tan \delta$	$Q_s = q_s \cdot A_s = \alpha \cdot C_u \cdot A_s$
$Q_b = q_b \cdot A_b = N_q \cdot \sigma'_{v0} \cdot A_b$	$Q_b = q_b \cdot A_b = N_c \cdot C_{ub} \cdot A_b$

Passive and active pressures:

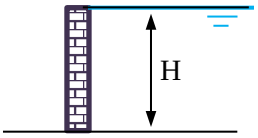
$\sigma'_h = K_a \cdot \sigma'_v$ (active)	$\sigma'_h = K_b \cdot \sigma'_v$ (passive)

continued

RANKINE'S theory: H high of the layer of soil behind the wall retaining wall, φ' is the friction angle and c' is the cohesion

In the following, $P_a = P_a' + U$ and $P_p = P_p' + U$,

where U is the resultant of the pore water pressure.



$K_a = \tan^2\left(\frac{\pi}{4} - \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1+\sin\varphi'}\right)^2 = \frac{1-\sin\varphi'}{1+\sin\varphi'}$	$K_p = \tan^2\left(\frac{\pi}{4} + \frac{\varphi'}{2}\right) = \left(\frac{\cos\varphi'}{1-\sin\varphi'}\right)^2 = \frac{1+\sin\varphi'}{1-\sin\varphi'}$	
Cohesionless soils:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u$	$\sigma_h' = \sigma_v' K_a - 2c'\sqrt{K_a}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{1}{2}(\gamma H - 2C_u)\left(H - \frac{2C_u}{\gamma}\right)$	$P_a' = \frac{K_a \gamma'}{2}\left(H - \frac{2c'}{\gamma'\sqrt{K_a}}\right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = 2C_u + \sigma_v$	$\sigma_h' = \sigma_v' K_p + 2c'\sqrt{K_p}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}(\gamma H + 4C_u)$	$P_p' = \frac{1}{2} K_p \gamma' H^2 + 2c'\sqrt{K_p} H$

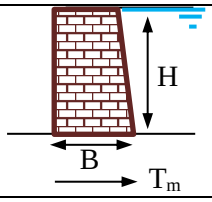
COULOMB'S theory: Friction and adhesion between soil and wall are φ_w and c_w, resp.

$K_a = \left(\frac{\cos\varphi'}{1+\sin\varphi'\sqrt{1+\frac{\tan\varphi_w'}{\tan\varphi'}}}\right)^2 \frac{1}{\cos\varphi_w'}$	$K_p = \left(\frac{\cos\varphi'}{1-\sin\varphi'\sqrt{1+\frac{\tan\varphi_w'}{\tan\varphi'}}}\right)^2 \frac{1}{\cos\varphi_w'}$	
Cohesionless:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u\sqrt{1+\frac{C_w}{C_u}}$	$\sigma_h' = K_{a1}\sigma_v' - c' K_{a2}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{\gamma}{2}\left(H - \frac{2C_u}{\gamma}\sqrt{1+\frac{C_w}{C_u}}\right)^2$	$P_a' = \frac{K_{a1}\gamma'}{2}\left(H - \frac{c' K_{a2}}{\gamma' K_{a1}}\right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = \gamma z + 2C_u\sqrt{1+\frac{C_w}{C_u}}$	$\sigma_h' = \gamma z K_{p1} + c' K_{p2}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}\left(\gamma H + 4C_u\sqrt{1+\frac{C_w}{C_u}}\right)$	$P_p' = \frac{1}{2} K_{p1}\gamma' H^2 + c' K_{p2} H$

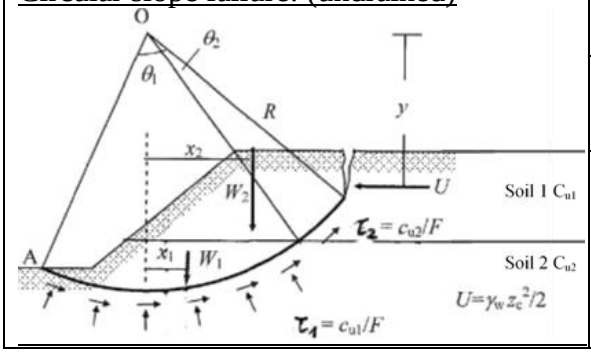
where, $K_{a1} = K_a$, $K_{p1} = K_p$ and

$K_{a2} = \frac{\cos\varphi'\left(2\sqrt{1+c_w'/c'} + (2+c_w'/c')\sin\varphi'\right)}{\left(1+\sin\varphi'\sqrt{1+\frac{\tan\varphi_w'}{\tan\varphi'}}\right)^2\cos\varphi_w'}$	$K_{p2} = \frac{\cos\varphi'\left(2\sqrt{1+c_w'/c'} - (2+c_w'/c')\sin\varphi'\right)}{\left(1-\sin\varphi'\sqrt{1+\frac{\tan\varphi_w'}{\tan\varphi'}}\right)^2\cos\varphi_w'}$
Cantilever wall in sand:	$FoS = \frac{N'\tan\varphi_w'}{T_m} = \frac{(W-U)\tan\varphi_w'}{T_m}$
Gravity wall in sand:	$FoS = \frac{N\tan\varphi_w'}{T_m}$

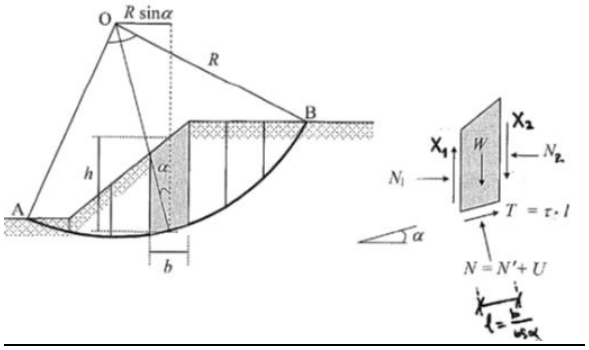
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Gravity wall in clay: 	(Undrained) $FoS = \frac{C_w B}{T_m}$	(Drained) $FoS = \frac{N'\tan\varphi_w' + c_w' B}{T_m}$

Slope stability: (Note: for undrained strength in infinite slope take c'=C_u and φ'=0)

Infinite slope with parallel flow: H: layer thickness, m: position of water table from bedrock, β: slope angle; γ_1 (dry soil; m=0), γ_2 (saturated soil; m=1).	$FoS = \frac{c' + H\left(\gamma_1(1-m) + m(\gamma_2 - \gamma_w)\right)\cos^2\beta\tan\varphi'}{H\left((1-m)\gamma_1 + m\gamma_2\right)\cos\beta\sin\beta}$
Vertical cut (dry soil):	$FoS = \frac{4C_u}{H\gamma\sin 2\alpha}$ (undrained)
	$FoS = \frac{4\left(c' + \sigma'\tan\varphi'\right)}{H\gamma\sin 2\alpha}$ (drained)
Circular slope failure: (undrained) 	$FoS = \frac{R^2\theta C_u}{Wx}$
	$FoS = \frac{R^2\theta C_u}{Wx + Uy}$ (with cracks)
	$FoS = \frac{\left(\theta_1 C_{u1} + \theta_2 C_{u2}\right)R^2}{W_1 x_1 + W_2 x_2 + Uy}$ (layered soil)

$\sin 2\alpha = 2\sin\alpha\cos\alpha$, $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$ and arc-length: $AB = R\theta$ (θ is in radians).

Circular: (drained): 	$FoS = \frac{\sum_{i=1}^{n} l\left(c' + \sigma'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} = \frac{\sum_{i=1}^{n}\left(lc' + N'\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$
Modified Bishop's b: width of the slice, l: length of the slice, α: inclination of the slice to the horizontal direction.	$FoS = \frac{\sum_{i=1}^{n}\left(bc' + \left(W - U\cos\alpha\right)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i} M(\alpha)$ where, $M(\alpha) = \frac{1}{\cos\alpha\left[1 + \frac{\tan\alpha\tan\varphi'}{FoS}\right]}$
Fellenius:	$FoS = \frac{\sum_{i=1}^{n}\left(lc' + \left(W\cos\alpha - ul\right)\tan\varphi'\right)_i}{\sum_{i=1}^{n}\left(W\sin\alpha\right)_i}$

END